

DBMS — Lecture 1

Introduction to DBMS & Historical Perspective

1. Learning Objectives

Understand data, database, and DBMS.

Understand the historical move from file-processing systems to database systems.

Identify why organizations need a DBMS.

2. Basic Concepts

Data: Raw facts and figures, such as 101, Raju, CSE, 85.

Database: An organized collection of related data that can be accessed, managed, and updated efficiently.

DBMS: Software used to create, store, organize, retrieve, update, and manage data in a database.

3. Historical Perspective

File Processing System: Earlier applications stored information in separate files. Each application often maintained its own files.

Database Approach: DBMS emerged to provide centralized management, controlled sharing, security, integrity, backup, recovery, and efficient access to related data.

Typical progression: **File Processing** → **Database Systems** → **Modern DBMS**.

4. Example — College Database

A college may maintain Student, Faculty, Course, Fees, Attendance, and Examination information. A DBMS can manage these related data items in an integrated database rather than isolated application files.

5. Why DBMS?

Traditional files can cause duplication, inconsistency, difficult sharing, weak security, and difficult recovery. DBMS provides systematic management of data and controlled access for multiple users.

Key Takeaway

DBMS is software that provides an organized, secure, and efficient way to store and manage data.